

13.3 Derivative. Analytic Function

Just as the study of calculus or real analysis required concepts such as domain, neighborhood, function, limit, continuity, derivative, etc., so does the study of complex analysis. Since the functions live in the complex plane, the concepts are slightly more difficult or *different* from those in real analysis. This section can be seen as a reference section where many of the concepts needed for the rest of Part D are introduced.

Circles and Disks. Half-Planes

The **unit circle** $|z| = 1$ (Fig. 330) has already occurred in Sec. 13.2. Figure 331 shows a general circle of radius ρ and center a . Its equation is

$$|z - a| = \rho$$

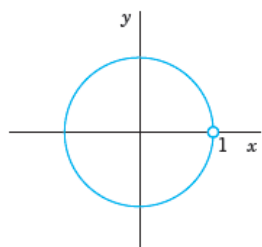


Fig. 330. Unit circle

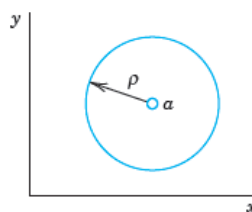


Fig. 331. Circle in the complex plane

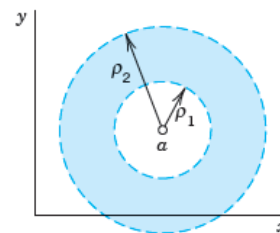


Fig. 332. Annulus in the complex plane

because it is the set of all z whose distance $|z - a|$ from the center a equals ρ . Accordingly, its interior (“**open circular disk**”) is given by $|z - a| < \rho$, its interior plus the circle itself (“**closed circular disk**”) by $|z - a| \leq \rho$, and its exterior by $|z - a| > \rho$. As an example, sketch this for $a = 1 + i$ and $\rho = 2$, to make sure that you understand these inequalities.

An open circular disk $|z - a| < \rho$ is also called a **neighborhood** of a or, more precisely, a ρ -neighborhood of a . And a has infinitely many of them, one for each value of ρ (> 0), and a is a point of each of them, by definition.

In modern literature *any set* containing a ρ -neighborhood of a is also called a *neighborhood* of a .

Figure 332 shows an **open annulus** (circular ring) $\rho_1 < |z - a| < \rho_2$, which we shall need later. This is the set of all z whose distance $|z - a|$ from a is greater than ρ_1 but less than ρ_2 . Similarly, the **closed annulus** $\rho_1 \leq |z - a| \leq \rho_2$ includes the two circles.

Half-Planes. By the (open) **upper half-plane** we mean the set of all points $z = x + iy$ such that $y > 0$. Similarly, the condition $y < 0$ defines the *lower half-plane*, $x > 0$ the *right half-plane*, and $x < 0$ the *left half-plane*.

Complex Function

Complex analysis is concerned with complex functions that are differentiable in some domain. Hence we should first say what we mean by a complex function and then define the concepts of limit and derivative in complex. This discussion will be similar to that in calculus. Nevertheless it needs great attention because it will show interesting basic differences between real and complex calculus.

Recall from calculus that a *real* function f defined on a set S of real numbers (usually an interval) is a rule that assigns to every x in S a real number $f(x)$, called the *value* of f at x . Now in complex, S is a set of *complex* numbers. And a **function** f defined on S is a rule that assigns to every z in S a complex number w , called the *value* of f at z . We write

$$w = f(z).$$

Here z varies in S and is called a **complex variable**. The set S is called the *domain of definition* of f or, briefly, the **domain** of f . (In most cases S will be open and connected, thus a domain as defined just before.)

Example: $w = f(z) = z^2 + 3z$ is a complex function defined for all z ; that is, its domain S is the whole complex plane.

The set of all values of a function f is called the **range** of f .

w is complex, and we write $w = u + iv$, where u and v are the real and imaginary parts, respectively. Now w depends on $z = x + iy$. Hence u becomes a real function of x and y , and so does v . We may thus write

$$w = f(z) = u(x, y) + iv(x, y).$$

This shows that a *complex* function $f(z)$ is equivalent to a *pair* of *real* functions $u(x, y)$ and $v(x, y)$, each depending on the two real variables x and y .

EXAMPLE 1 Function of a Complex Variable

Let $w = f(z) = z^2 + 3z$. Find u and v and calculate the value of f at $z = 1 + 3i$.

Solution. $u = \operatorname{Re} f(z) = x^2 - y^2 + 3x$ and $v = 2xy + 3y$. Also,

$$f(1 + 3i) = (1 + 3i)^2 + 3(1 + 3i) = 1 - 9 + 6i + 3 + 9i = -5 + 15i.$$

This shows that $u(1, 3) = -5$ and $v(1, 3) = 15$. Check this by using the expressions for u and v . ■

EXAMPLE 2 Function of a Complex Variable

Let $w = f(z) = 2iz + 6\bar{z}$. Find u and v and the value of f at $z = \frac{1}{2} + 4i$.

Solution. $f(z) = 2i(x + iy) + 6(x - iy)$ gives $u(x, y) = 6x - 2y$ and $v(x, y) = 2x - 6y$. Also,

$$f\left(\frac{1}{2} + 4i\right) = 2i\left(\frac{1}{2} + 4i\right) + 6\left(\frac{1}{2} - 4i\right) = i - 8 + 3 - 24i = -5 - 23i.$$

Check this as in Example 1. ■

Remarks on Notation and Terminology

1. Strictly speaking, $f(z)$ denotes the value of f at z , but it is a convenient abuse of language to talk about *the function* $f(z)$ (instead of *the function* f), thereby exhibiting the notation for the independent variable.

2. We assume all functions to be *single valued relations*, as usual: to each z in S there corresponds but *one* value $w = f(z)$ (but, of course, several z may give the same value $w = f(z)$, just as in calculus). Accordingly, we shall *not use* the term “multivalued function” (used in some books on complex analysis) for a multivalued relation, in which to a z there corresponds more than one w .

The limit of a function

The limit of $w = f(z)$ as $z \rightarrow z_0$ is a number ℓ such that $|f(z) - \ell|$ can be made as small as we wish by making $|z - z_0|$ sufficiently small. In some cases the limit is simply $f(z_0)$, as is the case for $w = z^2 - z$. For example, the limit of this function as $z \rightarrow i$ is $f(i) = i^2 - i = -1 - i$.

There is a fundamental difference from functions of a real variable: there, we could approach a point on the curve $y = g(x)$ either from the left or from the right when considering limits of $g(x)$ at such points. With the function $f(z)$ we are allowed to approach the point $z = z_0$ along *any* path in the z -plane; we require merely that the distance $|z - z_0|$ decreases to zero.

Suppose that we want to find the limit of $f(z) = z^2 - z$ as $z \rightarrow 2 + i$ along each of the paths (a), (b) and (c) indicated in Figure 1.

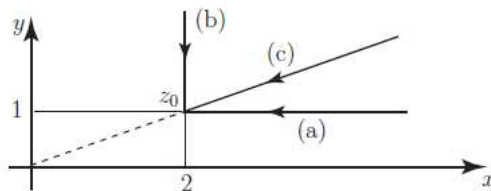


Figure 1

(a) Along this path $z = x + i$ (for any x) and $z^2 - z = x^2 + 2xi - 1 - x - i$

That is: $z^2 - z = x^2 - 1 - x + (2x - 1)i$.

As $z \rightarrow 2 + i$, then $x \rightarrow 2$ so that the limit of $z^2 - z$ is $2^2 - 1 - 2 + (4 - 1)i = 1 + 3i$.

(b) Here $z = 2 + yi$ (for any y) so that $z^2 - z = 4 - y^2 - 2 + (4y - y)i$.

As $z \rightarrow 2 + i$, $y \rightarrow 1$ so that the limit of $z^2 - z$ is $4 - 1 - 2 + (4 - 1)i = 1 + 3i$.

(c) Here $z = k(2 + i)$ where k is a real number. Then

$$z^2 - z = k^2(4 + 4i - 1) - k(2 + i) = 3k^2 - 2k + (4k^2 - k)i.$$

As $z \rightarrow 2 + i, k \rightarrow 1$ so that the limit of $z^2 - z$ is $3 - 2 + (4 - 1)i = 1 + 3i$.

In each case the limit is the same.

Evaluate the limit of $f(z) = z^2 + z + 1$ as $z \rightarrow 1 + 2i$ along the paths

- (a) parallel to the x -axis coming from the right,
- (b) parallel to the y -axis, coming from above,
- (c) the line joining the point $1 + 2i$ to the origin, coming from the origin.

Answer

- (a) Along this path $z = x + 2i$ and $z^2 + z + 1 = x^2 - 4 + x + 1 + (4x + 2)i$. As $z \rightarrow 1 + 2i, x \rightarrow 1$ and $z^2 + z + 1 \rightarrow -1 + 6i$.
- (b) Along this path $z = 1 + yi$ and $z^2 + z + 1 = 1 - y^2 + 1 + 1 + (2y + y)i$. As $z \rightarrow 1 + 2i, y \rightarrow 2$ and $z^2 + z + 1 \rightarrow -1 + 6i$.
- (c) If $z = k(1 + 2i)$ then $z^2 + z + 1 = k^2 + k + 1 - 4k^2 + (4k^2 + 2k)i$. As $z \rightarrow 1 + 2i, k \rightarrow 1$ and $z^2 + z + 1 \rightarrow -1 + 6i$.

Not all functions of a complex variable are as straightforward to analyse as the last two examples. Consider the function $f(z) = \frac{\bar{z}}{z}$. Along the x -axis moving towards the origin from the right

$z = x$ and $\bar{z} = x$ so that $f(z) = 1$ which is the limit as $z \rightarrow 0$ **along this path**.

However, we can approach the origin along any path. If instead we approach the origin along the positive y -axis $z = iy$ then

$\bar{z} = -iy$ and $f(z) = \frac{\bar{z}}{z} = -1$, which is the limit as $z \rightarrow 0$ **along this path**.

Since these two limits are distinct then $\lim_{z \rightarrow 0} \frac{\bar{z}}{z}$ **does not exist**.

We cannot assume that the limit of a function $f(z)$ as $z \rightarrow z_0$ is independent of the path chosen.

Definition of continuity

The function $f(z)$ is **continuous** as $z \rightarrow z_0$ if the following two statements are true:

- (a) $f(z_0)$ exists;
- (b) $\lim_{z \rightarrow z_0} f(z)$ exists and is equal to $f(z_0)$.

As an example consider $f(z) = \frac{z^2 + 4}{z^2 + 9}$. As $z \rightarrow i$, then $f(z) \rightarrow f(i) = \frac{i^2 + 4}{i^2 + 9} = \frac{3}{8}$. Thus $f(z)$ is continuous at $z = i$.

However, when $z^2 + 9 = 0$ then $z = \pm 3i$ and neither $f(3i)$ nor $f(-3i)$ exists. Thus $\frac{z^2 + 4}{z^2 + 9}$ is discontinuous at $z = \pm 3i$. It is easily shown that these are the *only* points of discontinuity.

State where $f(z) = \frac{z}{z^2 + 4}$ is discontinuous. Find $\lim_{z \rightarrow i} f(z)$.

Answer

$z^2 + 4 = 0$ where $z = \pm 2i$; at these points $f(z)$ is discontinuous as $f(\pm 2i)$ does not exist.

$$\lim_{z \rightarrow i} f(z) = f(i) = \frac{i}{i^2 + 4} = \frac{1}{3}i.$$

It is easily shown that any polynomial in z is continuous everywhere whilst any rational function is continuous everywhere except at the zeroes of the denominator.

Derivative

The **derivative** of a complex function f at a point z_0 is written $f'(z_0)$ and is defined by

1

$$f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

provided this limit exists. Then f is said to be **differentiable** at z_0 . If we write $\Delta z = z - z_0$, we have $z = z_0 + \Delta z$ and (1) takes the form

2

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}.$$

Now comes an *important point*. Remember that, by the definition of limit, $f(z)$ is defined in a neighborhood of z_0 and z in (2) may approach z_0 from any direction in the complex plane. Hence differentiability at z_0 means that, along whatever path z approaches z_0 , the quotient in (2) always approaches a certain value and all these values are equal. This is important and should be kept in mind.

EXAMPLE 3 Differentiability. Derivative

The function $f(z) = z^2$ is differentiable for all z and has the derivative $f'(z) = 2z$ because

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{(z + \Delta z)^2 - z^2}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{z^2 + 2z \Delta z + (\Delta z)^2 - z^2}{\Delta z} = \lim_{\Delta z \rightarrow 0} (2z + \Delta z) = 2z. \quad \blacksquare$$

The **differentiation rules** are the same as in real calculus, since their proofs are literally the same. Thus for any differentiable functions f and g and constant c we have

$$(cf)' = cf', \quad (f + g)' = f' + g', \quad (fg)' = f'g + fg', \quad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

as well as the chain rule and the power rule $(z^n)' = nz^{n-1}$ (n integer).

Also, if $f(z)$ is differentiable at z_0 , it is continuous at z_0 . (See Team Project 24.)

EXAMPLE 4 $\bar{z}$ not Differentiable

It may come as a surprise that there are many complex functions that do not have a derivative at any point. For instance, $f(z) = \bar{z} = x - iy$ is such a function. To see this, we write $\Delta z = \Delta x + i\Delta y$ and obtain

$$(5) \quad \frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{\overline{(z + \Delta z)} - \bar{z}}{\Delta z} = \frac{\overline{\Delta z}}{\Delta z} = \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y}.$$

If $\Delta y = 0$, this is $+1$. If $\Delta x = 0$, this is -1 . Thus (5) approaches $+1$ along path I in Fig. 334 but -1 along path II. Hence, by definition, the limit of (5) as $\Delta z \rightarrow 0$ does not exist at any z . ■

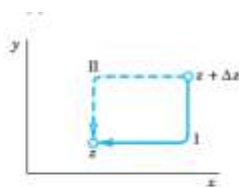


Fig. 334. Paths in (5)

Surprising as Example 4 may be, it merely illustrates that differentiability of a *complex* function is a rather severe requirement.

The idea of proof (approach of z from different directions) is basic and will be used again as the crucial argument in the next section.